FORWARD DEPLOYED / BACKEND SYSTEMS / CASE STUDY

INVARIANT OPS

CASE STUDY

05

Invariant Ops asks which patterns keep procurement records correct when integrations fail.

PROPOSED RESEARCH / NOT YET VERIFIED

THE BUSINESS CONTEXT

A mid sized distributor raises purchase orders, records goods receipts and approves supplier invoices across several systems that fail in ordinary ways. Messages arrive twice or late and a process can stop halfway. Wrong ledger entries are expensive to find. This case study uses a synthetic workflow only.

WHO IS INVOLVED AND WHAT THEY NEED

01	**Procurement manager**	Needs orders and receipts to agree without chasing teams.
02	**Finance operations lead**	Needs no duplicate postings and a clear way to correct errors.
03	**Platform engineer**	Needs failures to be replayable rather than mysterious.

SCOPE AND HONEST LIMITS

- The workflow is synthetic and does not replace evidence from real delivery.
- Exactly once outcomes depend on application rules and not middleware alone.
- No claim is made about production scale or measured business savings.
- Failure injection covers chosen scenarios and cannot cover every real fault.

CURRENT WORKFLOW AND FAILURE POINTS

05

PROPOSED RESEARCH / NOT YET VERIFIED

HOW THE WORK RUNS TODAY

1 TODAY
Order raised
A purchase order is created in one system and copied to others by script.

2 TODAY
Goods received
Receipts are keyed or imported and sometimes arrive before the order.

3 TODAY
Invoice arrives
Supplier invoices are matched to orders by spreadsheet and memory.

4 TODAY
Integration retries
Failed calls are retried by hand or by cron with no duplicate protection.

5 TODAY
Month end clean up
Finance finds mismatches and fixes ledgers through manual adjustments.

6 TODAY
Explain the gap
Engineers search logs to find why records disagree.

WHERE IT BREAKS

Duplicate postings	Retries and repeated messages create the same entry more than once.
Half finished work	A crash between steps leaves records in an unknown state.
Silent schema drift	A changed field format breaks a consumer without any clear alert.
Slow reconciliation	Differences are found late and resolved by hand.

REDESIGNED WORKFLOW AND DECISION AUTHORITY

05

PROPOSED RESEARCH / NOT YET VERIFIED

THE REDESIGNED WORKFLOW

1 SYSTEM
Record the event
Each business change is written with its outbox entry in one transaction.

→

2 SYSTEM
Deliver safely
Consumers apply each message once through idempotency keys and an inbox table.

→

3 SYSTEM
Enforce invariants
The domain workflow rejects any transition that breaks a stated rule.

↓

4 SYSTEM
Detect divergence
Reconciliation compares systems and raises exceptions with the evidence.

→

5 PERSON
Resolve exceptions
A finance user chooses the correction from the options shown.

→

6 PERSON
Review failure traces
An engineer reviews replayed failures and signs off any rule change.

SYSTEM STEP PERSON DECISION

WHERE AUTHORITY STAYS WITH A PERSON

Correct a mismatched record	FINANCE OPERATIONS LEAD	Changing financial records is consequential and needs a named owner.
Approve an invoice for payment	PROCUREMENT MANAGER	The system checks the match but a person accepts the commitment.
Change a business invariant	PLATFORM ENGINEER	Rules define correctness so changes need review and a recorded reason.

ARCHITECTURE, DATA AND CONTROLS

05

PROPOSED RESEARCH / NOT YET VERIFIED

SYSTEM ARCHITECTURE

1	**External systems**	Synthetic legacy interfaces for orders, receipts and invoices with injected faults.
2	**Durable events**	Transactional outbox and inbox with idempotent consumers on PostgreSQL.
3	**Domain workflow**	Purchase order, receipt and invoice state machines with explicit invariants.
4	**Reconciliation**	Scheduled comparison, exception queue and guided manual correction.
5	**Failure injection**	Deterministic crash, retry and schema change scenarios for replay.

CORE DATA ENTITIES

- **Purchase order**
- **Goods receipt**
- **Supplier invoice**
- **Outbox event**
- **Idempotency key**
- **Exception case**

RISKS AND THE CONTROL FOR EACH

Duplicate payment or posting →	Idempotency keys and unique constraints reject repeated effects.
Crash between commit and send →	Outbox entries commit with state and are sent again after restart.
Stale schema in a message →	Versioned contracts quarantine unknown formats for review.
Unreviewed manual correction →	Every correction records actor, reason and before and after values.

EVALUATION AND DELIVERY PLAN

05

PROPOSED RESEARCH / NOT YET VERIFIED

HOW SUCCESS IS MEASURED

Duplicate suppression	Duplicate effects observed under repeated retry and replay campaigns.
Reconciliation completeness	Share of injected divergences found and explained by reconciliation.
Recovery latency	Time from restart to a consistent state after each injected crash.
Backlog under failure	Queue depth and drain time while a dependency is unavailable.
Invariant violations	Count of broken business rules across all failure campaigns.

TEST PLAN

1. **Crash the process at commit, acknowledgement, retry and restart boundaries.**
2. **Replay the same message set several times and compare final ledger state.**
3. **Send messages with old and new schema versions in mixed order.**
4. **Run randomised failure campaigns from fixed seeds so each run can be repeated.**

DELIVERY PLAN

01 ALIGN	02 UNDERSTAND	03 DESIGN	04 DELIVER	05 LAUNCH
Agree the business invariants that must never break and who may change them.	Map how orders, receipts and invoices move today and where they diverge.	Define state machines, event contracts, reconciliation rules and role boundaries.	Build the workflow and failure harness and run the campaigns.	Run beside the existing process and reconcile both before any cutover.
DECISION Which invariants are in scope	**DECISION** Which failure modes to inject	**DECISION** Approve invariants and roles	**DECISION** Accept the test evidence	**DECISION** Whether to proceed past shadow

DELIVERABLES

- Executable invariant test suite
- Failure trace library with replay scripts
- Local deployment guide
- Reconciliation rules and exception playbook
- Architecture decision record